TASK 2 · DISEASE INVOLVEMENT & LITERATURE

Huntington’s disease and the occipital lobe

Group 1 · HD_ICM_random200 · Cuneus, lingual, and inferior, middle and superior occipital gyri (left and right)

Beyond the striatum: HD thins the visual cortex

CAUSE

Expanded CAG repeat in the HTT gene (chromosome 4); 40 or more repeats means full penetrance

SPREAD

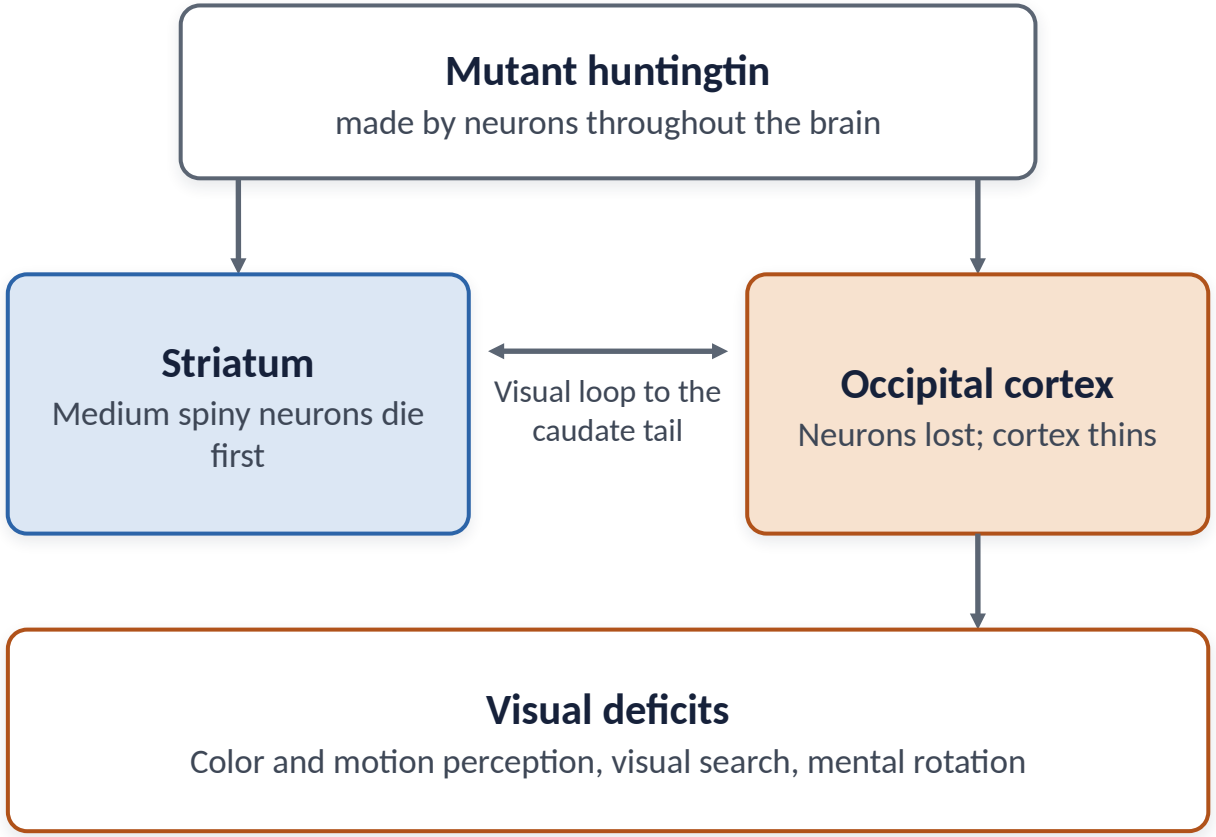
Striatum first; cortical thinning then moves from posterior to anterior regions

SYMPTOMS

Motor (chorea), cognitive and psychiatric; onset usually at age 30–50

SILENT PHASE

Brain changes begin 10–20 years before clinical diagnosis



Sources: HD Collaborative Research Group 1993; Paulsen 2008; Rosas 2002; Nana 2014; Seger 2013; Dhalla 2019

Studies place the occipital lobe early in HD

2002 · Rosas et al.

MRI · cortical thickness

Cortex thins early, spreading from posterior to anterior regions

2008 · Rosas et al.

MRI · stage I–II patients

Occipital and sensorimotor cortex already thin; anterior frontal spared

2014–15 · Nana; Rüb

Post-mortem neuron counts

Neurons lost across the cortex; data on primary visual cortex conflict (spared vs −32%)

2015 · Johnson et al.

TRACK-HD MRI · n = 275 · same study as our data

Occipital thinning even before symptoms; tracks worse visual-test scores

2016 · Labuschagne et al.

MRI + visual cognition

Visual search and mental rotation track posterior (parietal and occipital) volume

2021 · Johnson et al.

10-year longitudinal MRI

Occipital and parietal cortex atrophy fastest of all cortical regions

Full citations on the references slide

What we expect in our 10 occipital ROIs

Cu

Cuneus

Thins in HD, but thinning is not linked to visual-task scores

Johnson 2015

LG

Lingual gyrus

Thinning tracks worse visual cognition; weaker connectivity in manifest HD

Johnson 2015; Coppen 2018

IOG

Inferior occipital gyrus

Lateral occipital cortex: thinning tracks worse visual cognition

Johnson 2015

MOG

Middle occipital gyrus

Also lateral occipital; occipital cortex atrophies fastest over 10 years

Johnson 2015; 2021

SOG

Superior occipital gyrus

Dorsal-stream zone: posterior volume tracks visual search and mental rotation

Labuschagne 2016

Hypothesis for Task 4: HD subjects have smaller volumes than controls in all 10 occipital ROIs, left and right, with the lingual and lateral occipital gyri (IOG, MOG) most tied to symptoms.

FreeSurfer “lateral occipital” ≈ our IOG, MOG and part of SOG; calcarine V1 is split between our Cu and LG

References: disease involvement and literature

Coppen EM et al. (2018) Behav Brain Res 351:63–74

Coppen EM et al. (2018) Hum Brain Mapp 39:4776–4786

Dhalla A et al. (2019) J Huntingtons Dis 8:233–242

HD Collaborative Research Group (1993) Cell 72:971–983

HDSA. Overview of Huntington's disease. hdsa.org

Johnson EB et al. (2015) Neuropsychologia 79:138–146

Johnson EB et al. (2021) Biol Psychiatry 89:807–816

Labuschagne I et al. (2016) J Int Neuropsychol Soc 22:595–608

Nana AL et al. (2014) J Huntingtons Dis 3:45–64

Paulsen JS et al. (2008) J Neurol Neurosurg Psychiatry 79:874–880

Rosas HD et al. (2002) Neurology 58:695–701

Rosas HD et al. (2008) Brain 131:1057–1068

Rüb U et al. (2015) Brain Pathol 25:701–711

Seger CA (2013) Front Syst Neurosci 7:104

Tabrizi SJ et al. (2009) Lancet Neurol 8:791–801 (TRACK-HD)

Tabrizi SJ et al. (2020) Nat Rev Neurol 16:529–546